James Akintunde

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SUMMARY

Software engineer building AI agents at LSPedia for regulated pharma software. I design, test and cut the cost of LLM agents, and build full-stack apps in Python, TypeScript, React and SQL.

SKILLS

AI and LLM Engineering: AI agents, multi-agent orchestration, prompt engineering, RAG, embeddings and vector search, LLM evaluation and adversarial testing, guardrails, Model Context Protocol (MCP), token cost optimization; OpenAI, Anthropic Claude, Google Gemini, Azure AI Foundry

Software Engineering: Python, TypeScript, JavaScript, SQL, Java, C#; Node.js, Express, React, Vite, .NET Core, ASP.NET MVC; PostgreSQL, SQLite, Docker, BullMQ job queues, REST APIs, Git, GitHub, Azure

Testing and Domain: Playwright, Appium, Postman, Swagger, Zephyr Scale, regression, exploratory and accessibility testing; Jira, Confluence, Agile/Scrum; DSCSA, GS1, EPCIS, ANSI X12 EDI, GAMP 5

EXPERIENCE

Software Engineer 1, Research and Development

LSPedia **June 2026 - Present, Livonia, MI**

- **AI Agent Engineering:** Designed and built 4 LLM agents for pharmaceutical supply chain software (DSCSA, GS1, EPCIS) that draft requirements, review specs, turn Jira tickets into QA test plans and run manual tests.
- **Token Cost Optimization:** Cut a requirements agent's always-on prompt from about 4,500 tokens to 550, roughly 88% less, by splitting it into a small router with skills loaded on demand.
- **LLM Evaluation and Guardrails:** Set up agent evals with session compliance logs, a monthly 10-question knowledge check and adversarial tests for prompt injection, rule bypass and multi-intent requests.
- **RAG and LLM Platform:** Built a self-hosted chat platform in TypeScript, React and Node.js that streams from 4 LLM providers (OpenAI, Anthropic, Gemini, Azure AI Foundry), with a 4-tier RAG pipeline, RBAC and AES-256 key encryption.
- **Data Pipelines:** Built a multi-tenant EDI to EPCIS converter with React, PostgreSQL, BullMQ and Docker, adding DSCSA validation and 6-year data retention.
- **Quality Engineering:** Test feature branches for 4 product teams; found a critical permissions defect and 5 defects on a feature that passed every written requirement.

Culinary Specialist

U.S. Army Reserve **January 2025 - Present, Military**

- **Leadership:** Led a 7-person food service team preparing meals for 200+ soldiers per shift under tight time and safety standards.
- **Inventory Management:** Tracked and forecast food supplies, cutting waste by about 15%.
- **Quality and Safety:** Enforced sanitation and food safety protocols with zero failed inspections.

Software Developer Intern

Infoware Limited **May 2021 - October 2021, Lagos, Nigeria**

- **Full-Stack Development:** Developed two full-stack web applications using .NET Core, ASP.NET MVC5, HTML, CSS, JavaScript, and C#, delivering scalable and user-friendly solutions.
- **Collaboration & Development:** Partnered with senior developers to design and implement software features aligned with industry standards and best practices.
- **Leadership & Performance Optimization:** Led a team of 3 front-end developers to improve CRM system performance and usability, resulting in enhanced user experience and greater system efficiency.

EDUCATION

M.Sc. Computer Science (AI concentration)

Wayne State University · Detroit, MI · 2025 · 3.40/4

B.Sc. Computer Science

Mountain Top University · Ibafo, Ogun State, Nigeria · 2022 · 3.35/4

CERTIFICATIONS

Java and J2EE Developer
Edureka • 2023

Programming Foundations with JavaScript, HTML, and CSS
Duke University • 2022

Fundamentals of Network Communication
University of Colorado System • 2021

PROJECT

Multi-Agent Software Delivery System
Personal Project • September 2026 - Present
- Built a Claude Code plugin that divides software tasks across 5 specialized agents (planner, researcher, builder, reviewer, QA), each with its own role, tools and handoff format.
- Separated builder and reviewer agents so a different agent checks every change, and routed routine tasks to smaller models, cutting token cost by about 40%.
- Delivered 5+ web apps end to end with it, from research and planning to build and browser testing, saving screenshots and test results as proof for each task.

Driveway sealing app
School project • github.com/Burnout38x/driveway-sealing-app • January 2025 - December 2025
- Built a full-stack driveway sealing platform in .NET Core, JavaScript and SQL with 3 core modules (client sign-up, quotes, payments), tested against 50+ sample jobs.
- Designed audit tables that record 100% of quote and bill changes, giving every price update a full history.
- Added real-time scheduling and dispatch with Next.js and SQL triggers, cutting manual job assignment steps by about 40%.

Machine Learning for Healthcare Data (M.Sc. AI Coursework)
Wayne State University • March 2025 - March 2025
- Built and evaluated 4 supervised ML models in Python (pandas, scikit-learn) on a 10,000+ record healthcare dataset, covering data cleaning, feature engineering and 5-fold cross-validation.
- Compared models on precision, recall and F1, raising recall about 12% over the baseline, since a missed case carries clinical risk.
- Documented results and limits, including class imbalance and data quality, in 2 written reports for faculty review.